Table 1: **Phase 1 results for all models.** Benchmark columns are the endpoint after Tulu3 SFT→DPO, and the last three columns give the eleven-task suite average before Tulu3 and after each stage. MixtureVitae and Comma0.1 are permissive. Matched-compute models and SmolLM2 are at 16K context, Qwen2.5 and Qwen3 at their released context. Scores are on the full benchmarks, which include items identified in the training data (Section **??**), so they are not uniformly held-out estimates. Protocol A (Appendix Table **??** defines the abbreviations), means over three decoding seeds, with suite-average SD ≤ 0.7. Appendix Table 2 gives SDs and both context lengths.

	Mathematics					Code			Science/QA		IF	Suite average		
Model	A24	A25	AMC	M500	GSM8K	HE	LCB	MBPP	GPQA	JEE	IFE	Base	SFT	DPO
Matched compute, 300B tokens														
MixtureVitae	3.3	1.1	17.5	37.0	52.6	52.6	9.4	40.3	23.4	6.2	52.6	29.1	26.6	26.9
Comma0.1	0.0	0.0	2.5	2.1	6.7	17.1	0.0	25.9	23.4	5.1	48.8	11.1	11.7	12.0
Nemotron-CC-HQ	0.0	0.0	2.5	2.5	7.7	5.3	0.0	5.3	23.4	1.7	43.3	7.8	8.1	8.3
FineWeb-Edu	0.0	3.3	0.0	3.3	9.2	4.9	0.1	2.5	23.9	2.7	40.7	8.3	7.7	8.2
DCLM	0.0	0.0	0.0	1.4	7.1	6.1	0.0	4.9	25.8	1.1	44.1	7.5	8.1	8.2
External references, 11–36T tokens, uncontrolled														
SmolLM2 1.7B	0.0	0.0	3.3	9.6	33.8	32.3	3.2	36.9	26.3	5.4	54.9	16.8	18.2	18.7
Qwen2.5 1.5B	2.2	0.0	10.0	21.2	40.4	55.7	3.3	44.5	27.9	8.0	58.6	9.9	23.4	24.7
Qwen3 1.7B	0.0	0.0	8.3	31.2	69.8	67.7	4.1	54.7	31.3	13.3	64.9	30.7	31.5	31.4

Table 2: Phase 1 (Tulu3 SFT and DPO) for all models at the 1.7B scale, both context variants where both were evaluated. This is the complete version of Table 1. Protocol A (Appendix Table **??**). Values are mean$_{\pm SD}$ over decoding seeds 42, 1234, and 2025. The table has two panels: (a) mathematics, (b) code, science and instruction following, and the suite average.

(a) Mathematics

Model	**A24**	**A25**	**AMC**	**M500**	**GSM8K**
Matched compute, 300B tokens, controlled comparison					
MixtureVitae	$0.0_{\pm 0.00}$	$7.8_{\pm 5.09}$	$5.0_{\pm 0.00}$	$17.8_{\pm 0.72}$	$54.1_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$12.5_{\pm 0.00}$	$29.5_{\pm 0.23}$	$49.9_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$3.3_{\pm 0.00}$	$10.8_{\pm 1.44}$	$30.5_{\pm 0.12}$	$50.2_{\pm 0.00}$
MixtureVitae-16K	$1.1_{\pm 1.92}$	$3.3_{\pm 0.00}$	$23.3_{\pm 8.04}$	$47.0_{\pm 0.53}$	$60.0_{\pm 0.00}$
→ SFT	$3.3_{\pm 0.00}$	$3.3_{\pm 0.00}$	$15.0_{\pm 0.00}$	$35.0_{\pm 0.53}$	$51.6_{\pm 0.00}$
→ SFT → DPO	$3.3_{\pm 0.00}$	$1.1_{\pm 1.92}$	$17.5_{\pm 0.00}$	$37.0_{\pm 0.80}$	$52.6_{\pm 0.00}$
Nemotron-CC-HQ	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.2_{\pm 0.00}$	$4.1_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.0_{\pm 0.00}$	$3.1_{\pm 0.23}$	$6.8_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$3.5_{\pm 0.42}$	$8.0_{\pm 0.00}$
Nemotron-CC-HQ-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.8_{\pm 0.35}$	$8.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.9_{\pm 0.31}$	$8.6_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.5_{\pm 0.12}$	$7.7_{\pm 0.00}$
FineWeb-Edu	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.1_{\pm 0.12}$	$7.1_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$1.7_{\pm 0.31}$	$6.7_{\pm 0.00}$
FineWeb-Edu-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.0_{\pm 0.00}$	$1.8_{\pm 0.20}$	$8.1_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.1_{\pm 0.12}$	$8.7_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$3.3_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.3_{\pm 0.12}$	$9.2_{\pm 0.00}$
DCLM	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.2_{\pm 0.00}$	$3.0_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.8_{\pm 0.00}$	$5.2_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$1.1_{\pm 0.23}$	$4.6_{\pm 0.00}$
DCLM-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$1.5_{\pm 0.12}$	$5.8_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.8_{\pm 1.44}$	$1.3_{\pm 0.12}$	$7.2_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.4_{\pm 0.00}$	$7.1_{\pm 0.00}$
Comma0.1	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.5_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.23}$	$9.3_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.31}$	$9.7_{\pm 0.00}$
Comma0.1-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$2.8_{\pm 0.35}$	$12.2_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.5_{\pm 0.12}$	$6.9_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.00}$	$2.1_{\pm 0.12}$	$6.7_{\pm 0.00}$
External open-weights references, 11–36T tokens, uncontrolled					
SmolLM2	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$29.3_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 2.50}$	$6.3_{\pm 0.12}$	$33.3_{\pm 0.00}$
→ SFT → DPO	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$8.3_{\pm 0.46}$	$33.0_{\pm 0.00}$
SmolLM2-16K	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.8_{\pm 1.44}$	$11.8_{\pm 0.92}$	$31.9_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$9.6_{\pm 0.20}$	$32.8_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$3.3_{\pm 1.44}$	$9.6_{\pm 0.53}$	$33.8_{\pm 0.00}$
Qwen2.5	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.2_{\pm 0.35}$	$61.9_{\pm 0.00}$
→ SFT	$3.3_{\pm 0.00}$	$0.0_{\pm 0.00}$	$8.3_{\pm 1.44}$	$18.8_{\pm 0.69}$	$39.0_{\pm 0.00}$
→ SFT → DPO	$2.2_{\pm 1.92}$	$0.0_{\pm 0.00}$	$10.0_{\pm 2.50}$	$21.2_{\pm 1.00}$	$40.4_{\pm 0.00}$
Qwen3	$3.3_{\pm 0.00}$	$2.2_{\pm 1.92}$	$20.8_{\pm 3.82}$	$48.0_{\pm 0.35}$	$71.6_{\pm 0.00}$
→ SFT	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$16.7_{\pm 2.89}$	$28.2_{\pm 0.35}$	$69.4_{\pm 0.00}$
→ SFT → DPO	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$8.3_{\pm 1.44}$	$31.2_{\pm 1.06}$	$69.8_{\pm 0.00}$

Table 2: (continued) Panel (b), code, science and instruction following, and the suite average.

(b) Code, science and instruction following, and the suite average							
Model	**HE**	**LCB**	**MBPP**	**GPQA**	**JEE**	**IFE**	**Avg**
Matched compute, 300B tokens, controlled comparison							
MixtureVitae	$13.6_{\pm 0.93}$	$7.7_{\pm 0.16}$	$19.5_{\pm 0.12}$	$24.4_{\pm 0.77}$	$2.9_{\pm 0.51}$	$34.7_{\pm 0.13}$	$17.0_{\pm 0.38}$
→ SFT	$44.3_{\pm 0.70}$	$1.2_{\pm 0.16}$	$39.9_{\pm 0.23}$	$20.7_{\pm 0.87}$	$3.7_{\pm 0.55}$	$48.0_{\pm 0.08}$	$22.7_{\pm 0.14}$
→ SFT → DPO	$47.4_{\pm 0.35}$	$1.0_{\pm 0.16}$	$41.2_{\pm 0.20}$	$21.2_{\pm 0.00}$	$4.5_{\pm 0.34}$	$49.8_{\pm 0.08}$	$23.6_{\pm 0.17}$
MixtureVitae-16K	$54.1_{\pm 0.35}$	$11.5_{\pm 0.16}$	$40.1_{\pm 0.23}$	$23.4_{\pm 0.29}$	$8.0_{\pm 0.24}$	$47.7_{\pm 0.34}$	$29.1_{\pm 0.69}$
→ SFT	$54.1_{\pm 0.35}$	$9.7_{\pm 0.16}$	$40.4_{\pm 0.35}$	$23.4_{\pm 0.29}$	$5.1_{\pm 0.27}$	$51.9_{\pm 0.46}$	$26.6_{\pm 0.11}$
→ SFT → DPO	$52.6_{\pm 0.70}$	$9.4_{\pm 0.31}$	$40.3_{\pm 0.12}$	$23.4_{\pm 0.29}$	$6.2_{\pm 0.20}$	$52.6_{\pm 0.60}$	$26.9_{\pm 0.04}$
Nemotron-CC-HQ	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.4_{\pm 0.00}$	$5.7_{\pm 0.29}$	$0.0_{\pm 0.00}$	$29.2_{\pm 0.08}$	$3.8_{\pm 0.02}$
→ SFT	$4.9_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.5_{\pm 0.23}$	$20.2_{\pm 0.00}$	$2.0_{\pm 0.17}$	$38.1_{\pm 0.33}$	$7.8_{\pm 0.04}$
→ SFT → DPO	$6.1_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.1_{\pm 0.12}$	$21.2_{\pm 0.00}$	$2.7_{\pm 0.66}$	$39.3_{\pm 0.40}$	$8.0_{\pm 0.07}$
Nemotron-CC-HQ-16K	$1.8_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.7_{\pm 0.31}$	$22.7_{\pm 0.51}$	$5.6_{\pm 1.04}$	$39.0_{\pm 0.13}$	$7.8_{\pm 0.07}$
→ SFT	$5.3_{\pm 0.35}$	$0.0_{\pm 0.00}$	$5.9_{\pm 0.12}$	$22.2_{\pm 0.87}$	$1.2_{\pm 0.08}$	$42.9_{\pm 0.13}$	$8.1_{\pm 0.11}$
→ SFT → DPO	$5.3_{\pm 0.35}$	$0.0_{\pm 0.00}$	$5.3_{\pm 0.12}$	$23.4_{\pm 1.17}$	$1.7_{\pm 0.37}$	$43.3_{\pm 0.42}$	$8.3_{\pm 0.06}$
FineWeb-Edu	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$25.1_{\pm 0.58}$	$0.0_{\pm 0.00}$	$34.7_{\pm 0.00}$	$5.7_{\pm 0.05}$
→ SFT	$2.4_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.3_{\pm 0.23}$	$26.9_{\pm 0.58}$	$1.6_{\pm 0.46}$	$34.1_{\pm 0.00}$	$6.9_{\pm 0.08}$
→ SFT → DPO	$2.4_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.9_{\pm 0.12}$	$25.9_{\pm 0.29}$	$1.6_{\pm 0.17}$	$33.9_{\pm 0.45}$	$7.0_{\pm 0.06}$
FineWeb-Edu-16K	$3.7_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 0.23}$	$25.4_{\pm 0.29}$	$7.9_{\pm 1.02}$	$38.1_{\pm 0.65}$	$8.3_{\pm 0.10}$
→ SFT	$4.9_{\pm 0.00}$	$0.0_{\pm 0.00}$	$2.5_{\pm 0.23}$	$23.6_{\pm 0.58}$	$2.4_{\pm 0.40}$	$39.2_{\pm 0.27}$	$7.7_{\pm 0.07}$
→ SFT → DPO	$4.9_{\pm 0.00}$	$0.1_{\pm 0.16}$	$2.5_{\pm 0.12}$	$23.9_{\pm 0.29}$	$2.7_{\pm 0.54}$	$40.7_{\pm 0.53}$	$8.2_{\pm 0.07}$
DCLM	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$19.5_{\pm 0.77}$	$0.0_{\pm 0.00}$	$37.0_{\pm 0.08}$	$5.4_{\pm 0.08}$
→ SFT	$6.7_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.4_{\pm 0.35}$	$21.7_{\pm 0.00}$	$1.1_{\pm 0.49}$	$39.5_{\pm 0.23}$	$7.3_{\pm 0.02}$
→ SFT → DPO	$6.7_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.6_{\pm 0.00}$	$20.7_{\pm 0.00}$	$0.5_{\pm 0.03}$	$41.3_{\pm 0.08}$	$7.5_{\pm 0.15}$
DCLM-16K	$0.4_{\pm 0.35}$	$0.3_{\pm 0.00}$	$4.4_{\pm 0.00}$	$23.7_{\pm 0.00}$	$5.0_{\pm 0.16}$	$39.2_{\pm 0.15}$	$7.5_{\pm 0.13}$
→ SFT	$6.3_{\pm 0.35}$	$0.0_{\pm 0.00}$	$4.9_{\pm 0.12}$	$25.4_{\pm 0.29}$	$1.1_{\pm 0.12}$	$41.9_{\pm 0.53}$	$8.1_{\pm 0.19}$
→ SFT → DPO	$6.1_{\pm 0.00}$	$0.0_{\pm 0.00}$	$4.9_{\pm 0.12}$	$25.8_{\pm 0.51}$	$1.1_{\pm 0.56}$	$44.1_{\pm 0.45}$	$8.2_{\pm 0.08}$
Comma0.1	$7.9_{\pm 0.00}$	$0.0_{\pm 0.00}$	$8.8_{\pm 0.20}$	$20.2_{\pm 1.34}$	$0.0_{\pm 0.00}$	$38.4_{\pm 0.00}$	$7.3_{\pm 0.12}$
→ SFT	$15.2_{\pm 0.00}$	$0.2_{\pm 0.16}$	$22.5_{\pm 0.46}$	$25.6_{\pm 0.29}$	$3.9_{\pm 0.25}$	$42.4_{\pm 0.08}$	$11.1_{\pm 0.08}$
→ SFT → DPO	$15.2_{\pm 0.00}$	$0.6_{\pm 0.16}$	$22.7_{\pm 0.12}$	$26.8_{\pm 1.34}$	$3.9_{\pm 0.49}$	$44.7_{\pm 0.53}$	$11.5_{\pm 0.07}$
Comma0.1-16K	$13.4_{\pm 0.00}$	$0.0_{\pm 0.00}$	$16.0_{\pm 0.20}$	$25.8_{\pm 1.01}$	$7.8_{\pm 0.19}$	$41.9_{\pm 0.64}$	$11.1_{\pm 0.22}$
→ SFT	$16.5_{\pm 0.00}$	$0.2_{\pm 0.31}$	$25.2_{\pm 0.35}$	$23.2_{\pm 0.00}$	$4.8_{\pm 0.66}$	$47.2_{\pm 0.08}$	$11.7_{\pm 0.04}$
→ SFT → DPO	$17.1_{\pm 0.00}$	$0.0_{\pm 0.00}$	$25.9_{\pm 0.31}$	$23.4_{\pm 0.29}$	$5.1_{\pm 0.12}$	$48.8_{\pm 0.68}$	$12.0_{\pm 0.04}$
External open-weights references, 11–36T tokens, uncontrolled							
SmolLM2	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.9_{\pm 0.12}$	$15.7_{\pm 1.01}$	$0.0_{\pm 0.00}$	$31.5_{\pm 0.23}$	$7.1_{\pm 0.08}$
→ SFT	$30.9_{\pm 0.70}$	$1.6_{\pm 0.00}$	$37.2_{\pm 0.20}$	$24.7_{\pm 1.01}$	$3.7_{\pm 0.37}$	$49.5_{\pm 0.40}$	$17.2_{\pm 0.34}$
→ SFT → DPO	$30.9_{\pm 1.76}$	$1.7_{\pm 0.16}$	$37.7_{\pm 0.42}$	$25.8_{\pm 0.87}$	$4.4_{\pm 0.30}$	$50.4_{\pm 0.46}$	$17.7_{\pm 0.19}$
SmolLM2-16K	$24.0_{\pm 0.35}$	$0.1_{\pm 0.16}$	$36.5_{\pm 0.12}$	$24.6_{\pm 0.29}$	$7.6_{\pm 0.39}$	$47.4_{\pm 0.15}$	$16.8_{\pm 0.06}$
→ SFT	$32.9_{\pm 0.61}$	$3.1_{\pm 0.31}$	$37.5_{\pm 0.23}$	$24.9_{\pm 1.91}$	$4.7_{\pm 0.72}$	$52.9_{\pm 0.39}$	$18.2_{\pm 0.03}$
→ SFT → DPO	$32.3_{\pm 1.06}$	$3.2_{\pm 0.16}$	$36.9_{\pm 0.50}$	$26.3_{\pm 0.00}$	$5.4_{\pm 0.69}$	$54.9_{\pm 0.08}$	$18.7_{\pm 0.18}$
Qwen2.5	$5.7_{\pm 0.70}$	$0.0_{\pm 0.00}$	$0.1_{\pm 0.12}$	$11.1_{\pm 1.82}$	$0.3_{\pm 0.10}$	$29.9_{\pm 0.34}$	$9.9_{\pm 0.15}$
→ SFT	$51.0_{\pm 0.35}$	$3.6_{\pm 0.16}$	$44.3_{\pm 0.23}$	$29.0_{\pm 0.77}$	$6.1_{\pm 0.54}$	$54.0_{\pm 0.33}$	$23.4_{\pm 0.08}$
→ SFT → DPO	$55.7_{\pm 0.70}$	$3.3_{\pm 0.00}$	$44.5_{\pm 0.23}$	$27.9_{\pm 0.58}$	$8.0_{\pm 0.64}$	$58.6_{\pm 0.67}$	$24.7_{\pm 0.44}$
Qwen3	$64.2_{\pm 0.93}$	$1.8_{\pm 1.03}$	$46.5_{\pm 0.81}$	$25.1_{\pm 0.77}$	$18.0_{\pm 0.76}$	$36.4_{\pm 0.26}$	$30.7_{\pm 0.28}$
→ SFT	$67.1_{\pm 0.61}$	$4.4_{\pm 0.56}$	$54.3_{\pm 0.58}$	$33.0_{\pm 1.05}$	$11.9_{\pm 0.54}$	$61.4_{\pm 0.26}$	$31.5_{\pm 0.33}$
→ SFT → DPO	$67.7_{\pm 0.61}$	$4.1_{\pm 0.47}$	$54.7_{\pm 0.23}$	$31.3_{\pm 0.87}$	$13.3_{\pm 1.25}$	$64.9_{\pm 1.47}$	$31.4_{\pm 0.23}$